

PROVISIONAL PATENT APPLICATION

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Title of Invention

KV-CACHE COMPANION PROCESSING UNIT (KV-CPU):

A Closed-Loop AI-Native Memory Compute Architecture Providing Near-Memory Attention Score Reduction, Decode-Step-Aware Hardware Eviction, Request-Isolated KV Block Orchestration, and Hardware-Enforced Kernel Memory Tier Integration for Transformer Inference Acceleration

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1. TITLE OF THE INVENTION

KV-Cache Companion Processing Unit (KV-CPU): A Closed-Loop AI-Native Memory Compute Architecture Providing Near-Memory Attention Score Reduction, Decode-Step-Aware Hardware Eviction, Request-Isolated KV Block Orchestration, and Hardware-Enforced Kernel Memory Tier Integration for Transformer Inference Acceleration

2. FIELD OF THE INVENTION

The present invention relates to semiconductor integrated circuit design, computer memory systems, and hardware accelerators for artificial intelligence inference workloads. Specifically, it discloses a companion processing unit — the KV-CPU — that occupies a purpose-built intermediate tier in the memory hierarchy of large language model (LLM) inference servers. The KV-CPU implements a closed-loop KV block lifecycle system: a single hardware device that governs compute (attention score reduction), policy (eviction and prefetch), isolation (request-tagged block management), and OS integration (kernel memory tier and control plane). This closed-loop architecture — not any individual component in isolation — constitutes the principal inventive concept. The device targets both ARM-based and x86-based server platforms via a CXL 3.0 / PCIe 5.0 host interface.

3. BACKGROUND OF THE INVENTION

3.1 The KV-Cache Scaling Crisis in Autoregressive Inference

Modern transformer-based LLMs generate text autoregressively: at each decoding step t , the model requires access to key vectors $K[0..t-1]$ and value vectors $V[0..t-1]$ for all previously generated tokens across all N layers and H attention heads. The memory volume of this KV-cache grows linearly with context length L :

$$KV_memory(L) = 2 \times N \times H \times D \times L \times \text{sizeof}(dtype) \text{ [bytes]}$$

For a representative 70B-parameter model ($N=80$ layers, $H=64$ heads, $D=128$ head dimension, FP16), at $L=128,000$ tokens this evaluates to 320 GB per active inference request — vastly exceeding the 80 GB HBM capacity of a single NVIDIA H100. Even at $L=32,000$ tokens the KV-cache is 80 GB, consuming the entire HBM. The consequence is a fundamental trilemma: limit concurrency, limit context length, or spill KV blocks to slower memory at the cost of latency that

directly increases time-to-first-token and inter-token latency. No current hardware device solves this trilemma.

3.2 Quantified Failure of Existing Approaches

Approach	Max Context @ 80 GB HBM	KV Spill Latency	Root Cause Failure
Baseline (no offload)	~20k tokens	N/A — OOM	GPU HBM exhausted; request fails
vLLM PagedAttention (SW)	~32k tokens	200–600 μ s/block	CPU has no semantic view of decode step; LRU misses transformer access pattern; pipeline bubbles
CXL Type 3 passive expander	~128k tokens	150–300 ns read (CXL.mem)	Purely passive — zero compute; full KV block must traverse PCIe for every attention op
HBM-PIM (Samsung, SK Hynix)	GPU HBM only (no expansion)	N/A (no eviction)	GPU-attached only; stateless; no eviction; no capacity expansion; context limit unchanged
NVMe KV offload	Unlimited (impractical)	50,000–150,000 μ s	I/O latency dominates; inter-token latency becomes seconds

3.3 Why Prior Art Does Not Anticipate This Invention — Structured Analysis

The following table demonstrates that no single prior art reference, nor any combination thereof, teaches the closed-loop KV lifecycle system of the present invention:

Prior Art	What It Does	Critical Limitation	How KV-CPU Closed-Loop Differs
HBM-PIM (Samsung HBM-PIM 2.0, SK Hynix AiM)	Adds simple ALU inside HBM stack for GEMV — generic matrix-vector ops for weight access	GPU-die-attached; cannot expand capacity. Stateless — no eviction, no prefetch, no request tagging. Generic GEMV, not attention-specific. No interaction with eviction policy.	NMCE is not a generic GEMV unit — it computes attention scores coupled to KV block lifecycle. It is part of a closed loop: score result drives HEPC priority, HEPC drives RTBD tier field, RTBD drives NMCE block location lookup. No PIM device implements this feedback loop.
CXL Type 3 Memory Expanders (Samsung CMM-H, Micron CZ120)	Additional DRAM capacity via CXL.mem protocol	Zero compute. Full KV vectors must traverse CXL+PCIe to GPU for every attention computation. No eviction policy. No request tracking. No kernel semantic interface.	KV-CPU is not a memory expander with compute added — it is a policy engine where memory is the storage substrate. The HEPC and RTBD operate continuously on the memory contents, making placement decisions that no CXL expander firmware provides.
Software KV eviction (vLLM, SGLang, TRT-LLM)	CPU-managed paging with software LRU/LFU eviction; some systems use prefix caching	Software policy cannot observe decode-step timing at hardware speed. Eviction decisions require kernel syscalls (200–600 μ s). No hardware enforcement. Eviction	HEPC is fixed-function silicon. Eviction decisions execute in ~50 ns from step-advance MMIO signal. Zero CPU path. Hardware priority score directly controls DMA engines — no software in the enforcement loop.

		paces with GPU compute stream.	
Near-memory/in-memory compute (general PIM research)	Research demonstrating PIM for various workloads including some ML ops	General PIM proposals do not address: (a) transformer-specific attention semantics, (b) coupling of compute output to eviction policy, (c) multi-tenant request isolation, (d) decode-step-aware prefetch.	The inventive step is not "compute near memory" (known) but the specific coupling: NMCE output (attention score) is semantically linked to block importance, which flows into HEPC priority, which governs RTBD tier. This feedback is transformer-specific and not generic PIM.

4. SUMMARY OF THE INVENTION

The present invention provides a KV-Cache Companion Processing Unit (KV-CPU) — a purpose-built semiconductor device implementing a closed-loop KV block lifecycle system. The central inventive concept is the feedback loop between four co-integrated hardware components:

NMCE (compute) drives attention scores -> scores inform HEPC priority -> HEPC drives RTBD tier placement -> RTBD lookup enables NMCE to locate blocks -> loop repeats each decode step

This closed-loop is not present in any prior art. Its four pillars are each independently patentable and are claimed independently to survive individual prosecution even if other pillars face prior art objections:

PILLAR I — Near-Memory Attention Score Reduction NMCE computes dot-product attention scores using key vectors resident in on-device memory, returning only scalar score values. This is not generic PIM — it is attention-specific compute tightly coupled to the KV block lifecycle, meaningless without the RTBD block-location lookup that precedes it and the HEPC priority update that follows it.	PILLAR II — Decode-Step-Aware Hardware Eviction HEPC computes a four-component hardware priority score (R+F+S+D) in fixed-function silicon, triggered by GPU MMIO step-advance signal. The S (step-proximity) component is the key novelty — it requires knowledge of transformer decode semantics, not just memory access history. Eviction and prefetch execute asynchronously with zero CPU involvement.
PILLAR III — Request-Tagged Multi-Tenant KV Isolation RTBD maintains per-block hardware metadata (request_id, layer_idx, head_idx, token range, tier, prefix flag, reference count) in SRAM CAM.	PILLAR IV — Hardware Control Plane Kernel Interface The kernel driver is the hardware control plane — not a software invention. Each madvise call and io_uring opcode terminates in a hardware register write that

Hardware reference counting enables a single physical prefix block to serve N concurrent requests without data replication — saving $C \times \text{prefix_KV_GB}$ where C is request concurrency.

changes HEPC or RTBD silicon state. Without these writes, the hardware policy loop cannot function. The software interface exists solely to convey LLM-semantic information (decode step, block hotness) to hardware that has no other path to receive it.

5. DETAILED DESCRIPTION OF THE INVENTION

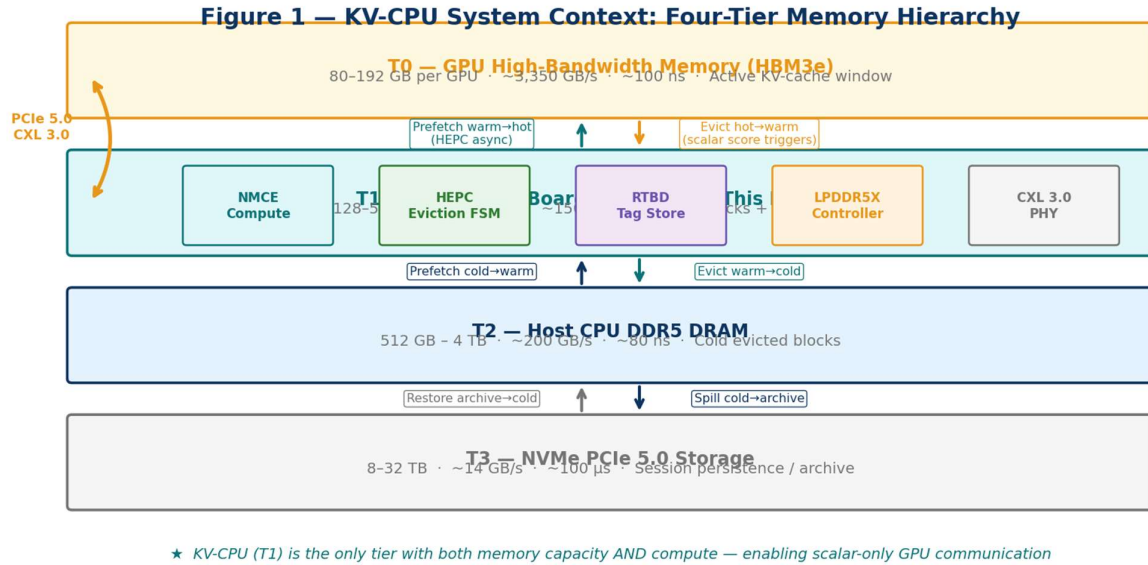


Figure 1 — KV-CPU System Context: Four-Tier Memory Hierarchy

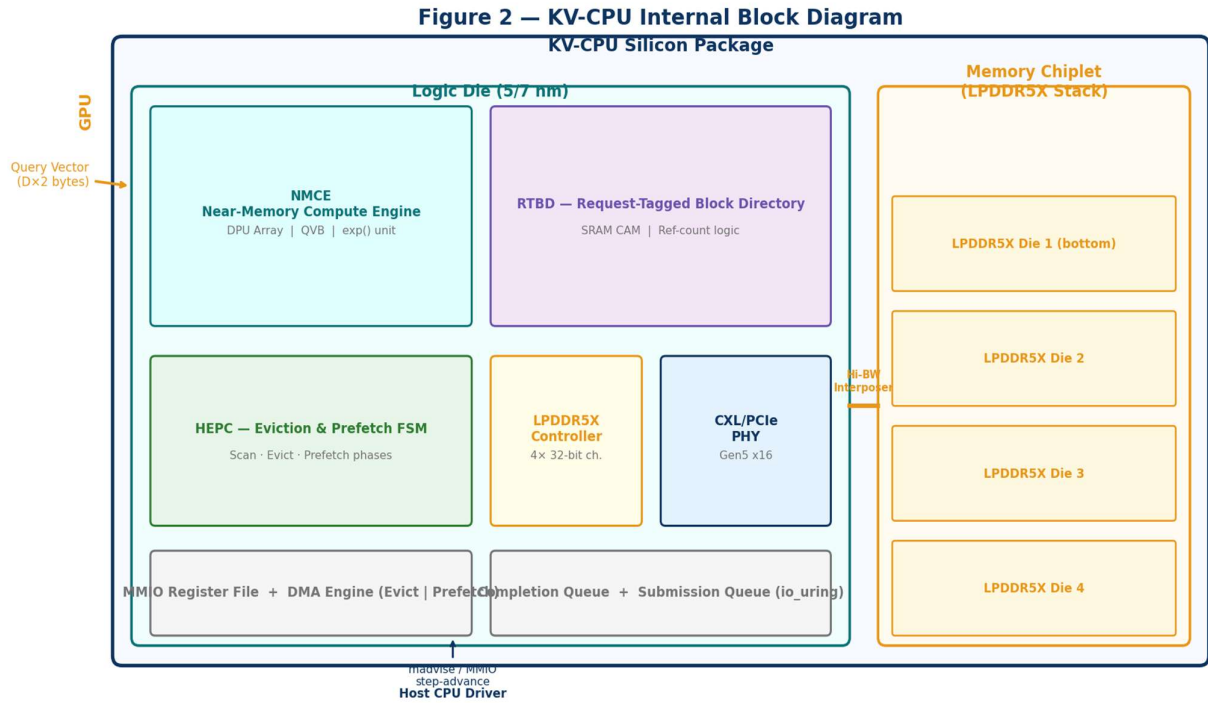
5.1 Architectural Context and Closed-Loop Operation

The KV-CPU implements a four-tier memory hierarchy as shown in Figure 1. The defining property of this hierarchy — absent in any prior art system — is that tier membership is determined not by the operating system's page replacement policy, nor by software running on a host CPU, but by a fixed-function hardware feedback loop operating continuously within the KV-CPU itself. Each decode step, the following cycle executes in hardware:

1. GPU writes current decode step t to HEPC step-advance register (MMIO write, ~50 ns)
2. HEPC SCAN: recomputes $P(B_i)$ for all tracked blocks in RTBD using recency R , frequency F , step-proximity S , prefix-dependency D
3. HEPC EVICT: schedules low-priority blocks for async DMA to T2/T3
4. HEPC PREFETCH: schedules predicted-needed blocks for async DMA from T2/T3 to T1
5. GPU issues attention compute descriptor for blocks at T1; NMCE computes scores from T1 LPDDR5X without key vector PCIe transfer

6. NMCE score results update per-block importance — implicitly feeding back into next SCAN phase priority

This closed loop runs entirely in silicon. No CPU instruction executes in the critical path between step-advance and completion of block scoring.



5.2 Pillar I — Near-Memory Compute Engine (NMCE): Attention-Specific Closed-Loop Compute

5.2.1 Why NMCE is Not Generic PIM

Processing-In-Memory (PIM) architectures such as Samsung HBM-PIM and SK Hynix AiM implement general-purpose GEMV operations. The NMCE is categorically different in three respects. First, its operation is semantically specific to scaled dot-product attention: the inputs are a query vector Q_h and a block of key vectors $K[0..B-1]$ already structured as a KV block — a data structure that only exists in transformer inference. Second, the output of the NMCE (B attention score scalars) is not the final result of a computation; it is a signal that drives the HEPC priority update for the block just scored. A generic PIM unit produces a result and stops. The NMCE produces a result that changes the hardware eviction policy for the very memory it just accessed. Third, the NMCE can only function correctly if the RTBD has correctly identified that the required block is resident in T1 and has provided its physical address — a lookup that is

meaningless without the RTBD's request-tagged block tracking. The NMCE is therefore not a separable compute unit but an integral component of the KV lifecycle system.

5.2.2 NMCE Operation and PCIe Traffic Reduction

When the GPU identifies that a KV block it needs is resident in T1, it issues an NMCE Compute Descriptor to the KV-CPU submission queue (CXL.io BAR) containing the query vector Q_h ($D \times 2$ bytes), the T1 address range of key block $K[0..B-1]$, and scaling factor $1/\sqrt{D}$. The NMCE executes entirely on-device:

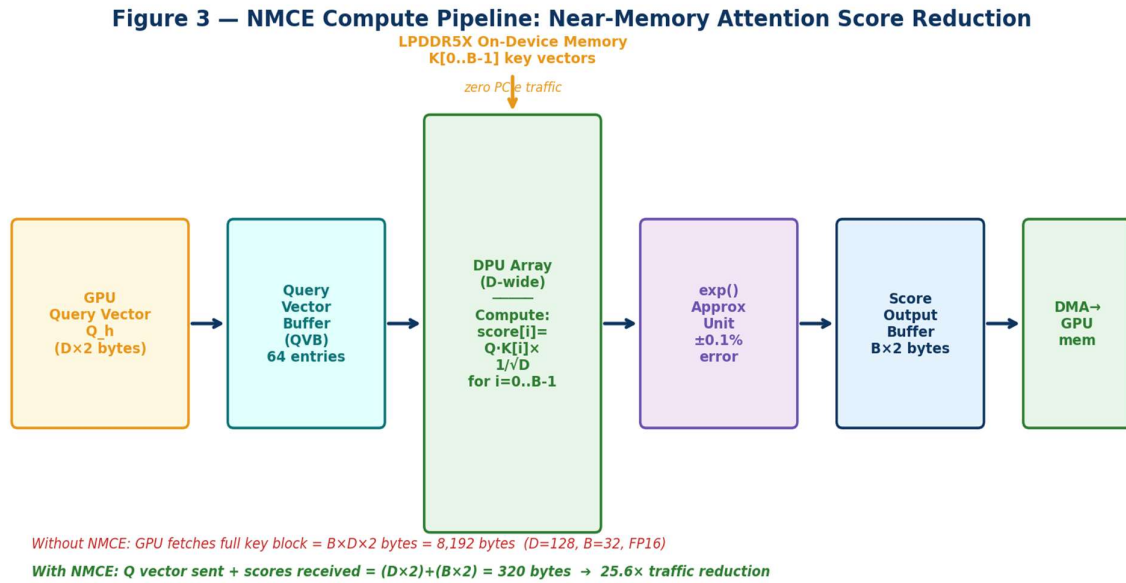
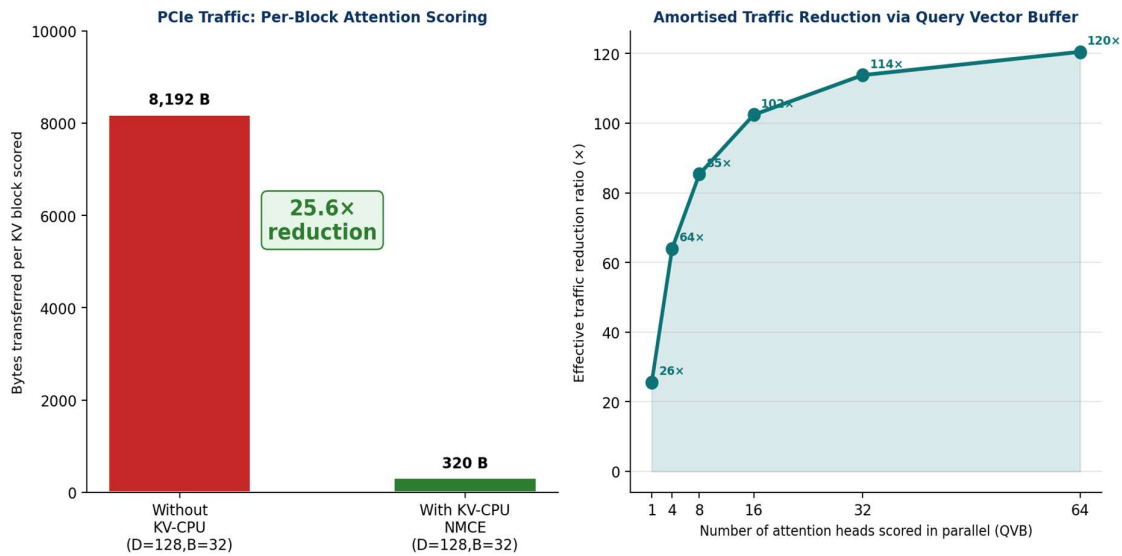


Figure 3 — NMCE Compute Pipeline: Attention-Specific Near-Memory Score Reduction

7. Fetches $K[0..B-1]$ from T1 LPDDR5X into local key line buffer (zero PCIe traffic)
8. Computes $\text{score}[i] = Q_h \cdot K[i] \times (1/\sqrt{D})$ for i in $[0, B-1]$ using D-wide DPU array
9. Applies piecewise-linear $\exp()$ approximation (8 segments, $\pm 0.1\%$ relative error)
10. Transmits B scalar score values ($B \times 2 = 64$ bytes for $B=32, FP16$) to GPU via CXL.io peer-to-peer write

Traffic reduction quantification ($D=128, B=32, FP16$): Without NMCE, GPU fetches full key block = $B \times D \times 2 = 8,192$ bytes. With NMCE, total exchange = $(D \times 2) + (B \times 2) = 320$ bytes. Ratio = 25.6x for single head; approaching 128x when 64 heads are amortised over the Query Vector Buffer (QVB) SRAM holding up to 64 simultaneous query vectors. See Figure 4.

Figure 4 — PCIe Traffic Comparison: KV Block Scoring Without vs With KV-CPU NMCE**Figure 4 — PCIe Traffic Comparison: Per-Block Scoring Without vs. With NMCE, and Amortised Multi-Head Reduction**

5.2.3 NMCE Hardware Components

- D-wide Dot-Product Unit (DPU) array: configurable D = 64/128/256, FP16/BF16 MAC
- Query Vector Buffer (QVB): 64-entry x D-element SRAM, amortises Q transmission across 64 attention heads
- Softmax $\exp()$ approximation unit: 8-segment piecewise linear, $\pm 0.1\%$ relative error
- Key Line Buffer: $2 \times B \times D \times 2$ byte SRAM ping-pong for double-buffered key prefetch from T1
- Completion DMA Engine: writes $B \times 2$ byte score vector to GPU-mapped BAR memory via CXL.io peer-to-peer

5.3 Pillar II — Hardware Eviction and Prefetch Controller (HEPC): Decode-Step-Aware Policy

5.3.1 The Decode-Step Awareness Problem and Why Software Cannot Solve It

Standard memory eviction policies (LRU, LFU, ARC, CLOCK) are access-history functions — they rank blocks by when and how often they were accessed. For transformer KV-cache, access history is insufficient because the access pattern is structurally determined by the decode step. As decoding progresses, the attention pattern across context positions changes in model-specific ways: system prompt blocks are accessed almost every step; recent-context blocks are accessed frequently; distant-context blocks may be dormant for hundreds of steps but will be re-accessed when the model's attention shifts. A hardware policy that observes memory

accesses but does not know the current decode step number cannot correctly distinguish "block is dormant because it is unimportant" from "block is dormant because the attention pattern has temporarily moved away from it." The HEPC's step-advance register solves this: it receives the current decode step t as a hardware signal from the GPU, making the hardware policy semantically aware of the inference session's temporal position.

Figure 5 — HEPC Three-Phase State Machine

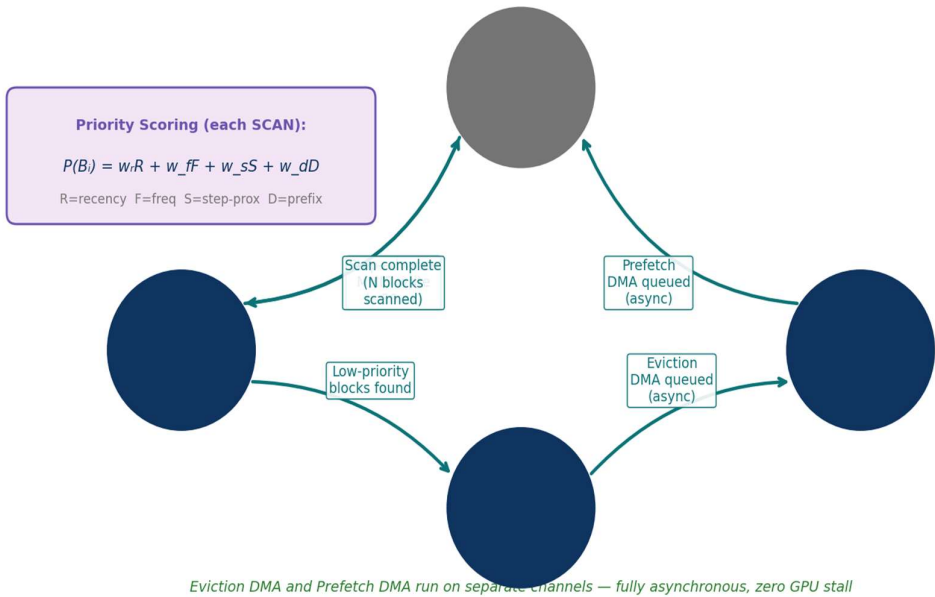


Figure 5 — HEPC Three-Phase State Machine: Scan, Evict, Prefetch (triggered by GPU MMIO step-advance)

5.3.2 Priority Scoring Function

The HEPC computes a composite priority score $P(B_i)$ for each tracked block B_i in hardware combinational logic during each SCAN phase:

$$P(B_i) = w_r \times R(B_i) + w_f \times F(B_i) + w_s \times S(B_i) + w_d \times D(B_i) \pmod{2^{16}}$$

Symbol	Component	Hardware Implementation	Why Transformer-Specific
$R(B_i)$	Recency	Saturating 8-bit counter reset to 255 on access; decremented by 1 per step-advance signal. Range 0–255.	Captures temporal locality within the decode session, not wall-clock time. Decay rate in decode steps, not seconds.
$F(B_i)$	Frequency	Saturating 8-bit counter, incremented on each block access. Range 0–255.	Captures working set membership — system prompt blocks hit every step quickly saturate to 255.

S(B _i)	Step proximity	$\max(0, W - (t_{\text{current}} - t_{\text{last_access}}))$. W = programmable window register.	The key novelty: measures recency in decode steps, not accesses. A block unaccessed for W steps is predicted outside current attention window. Generic PIM has no equivalent signal.
D(B _i)	Prefix dependency	Binary flag (is_prefix_block from RTBD). Weighted by w _d (default 200, max possible weight).	Shared prefix blocks must never be evicted while any request references them. D(B _i) combined with hardware reference count provides eviction guard — impossible in software at hardware speed.

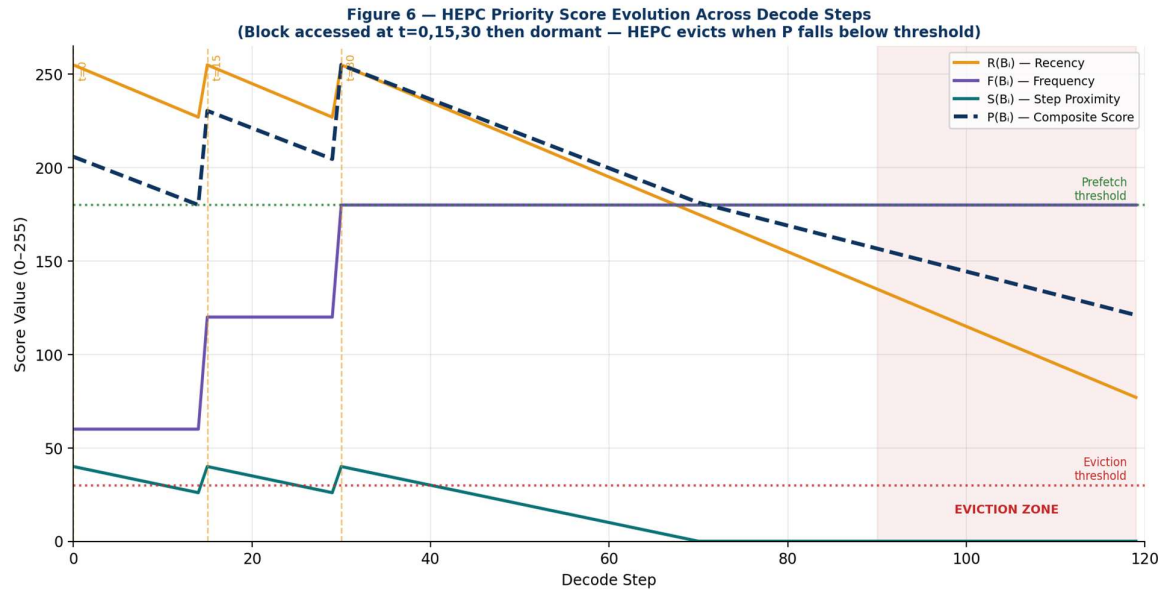


Figure 6 — Priority Score Component Evolution: Block accessed at t=0, 15, 30 then dormant; HEPC evicts when P falls below threshold

5.3.3 Three-Phase HEPC Cycle

11. SCAN PHASE: Triggered by GPU MMIO write to step-advance register (~50 ns). HEPC scans all active RTBD entries in parallel using pipelined comparator array, recomputing P(B_i) per block in one clock cycle each.
12. EVICT PHASE: Blocks with P(B_i) < eviction_threshold are placed in eviction DMA queue. Background DMA transfers to T2 (CXL.mem write) or T3 (io_uring IORING_OP_KV_EVICT). Separate from PREFETCH DMA channel — no serialization.
13. PREFETCH PHASE: Blocks in T2/T3 with P(B_i) > prefetch_threshold are staged into T1 LPDDR5X via background DMA. Triggered proactively before GPU requests — by definition the GPU never waits on a prefetched block.

5.4 Pillar III — Request-Tagged Block Directory (RTBD): Multi-Tenant Hardware Isolation

Figure 7 — RTBD Entry Bit-Field Layout (Total: ~240 bits / 30 bytes per entry)

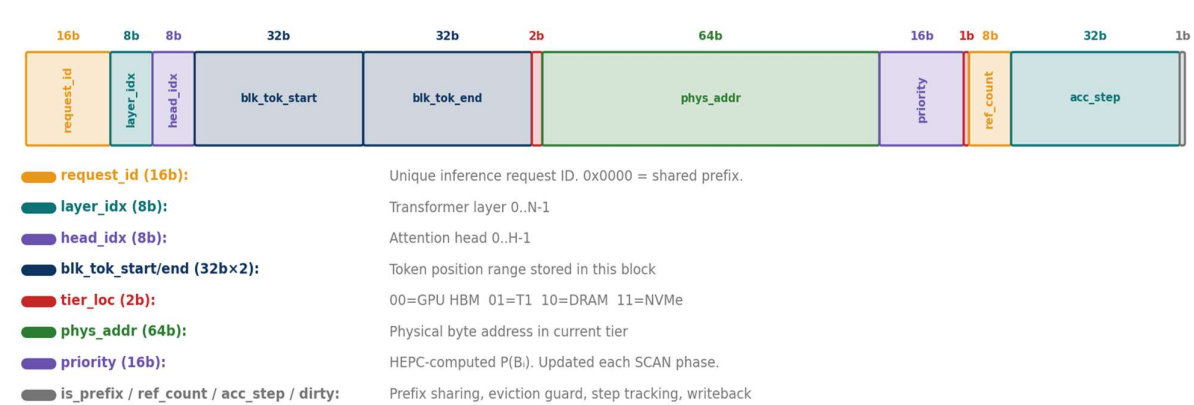


Figure 7 — RTBD Entry Bit-Field Layout: ~240 bits (30 bytes) per entry; up to 65,536 entries in SRAM CAM

5.4.1 Structure and Lookup

The RTBD is a fully-associative Content-Addressable Memory (CAM) structure in SRAM on the KV-CPU logic die. Each entry associates a physical KV block with: request_id (16 bits), layer_idx (8 bits), head_idx (8 bits), block_token_start and block_token_end (32 bits each), tier_location (2 bits: 00=GPU HBM, 01=T1, 10=T2, 11=T3), physical_addr (64 bits), priority_score (16 bits), is_prefix_block (1 bit), reference_count (8 bits), access_step (32 bits), dirty_bit (1 bit). Total: ~240 bits per entry. The CAM provides single-cycle lookup given a (request_id, layer_idx, head_idx, block_token_start) tuple — the query the GPU issues to locate a needed KV block.

5.4.2 Hardware Prefix Block Sharing via Reference Counting

In production LLM deployments, all requests from a single application share a common system prompt prefix. Without sharing, C concurrent requests consume C x prefix_KV_memory bytes of T1. The RTBD eliminates this waste:

Figure 8 — RTBD Prefix Block Sharing: Hardware Reference Counting
(Prefix length=4096 tokens, 70B model FP16 → 10.7 GB per prefix copy)

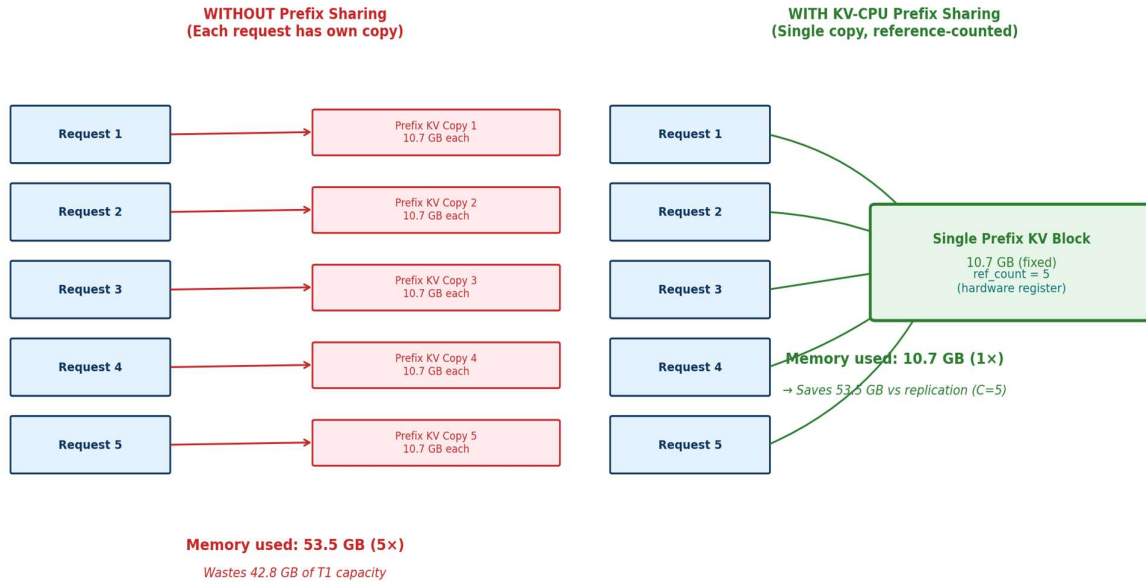


Figure 8 — Prefix Block Sharing: Hardware reference counting vs. replication (10 requests, 70B model, 4096-token prefix = 10.7 GB)

14. Prefix blocks registered with `is_prefix_block=1` and `request_id=0x0000` are stored as a single physical copy
15. Additional requests sharing the prefix issue `RTBD_SHARE` command (CXL.io register write), incrementing hardware `reference_count`
16. NMCE scores prefix blocks using the single physical copy — no data duplication
17. `is_prefix_block=1` AND `reference_count > 0` is a hardware eviction guard — HEPC cannot evict such blocks regardless of priority score
18. On request termination, `RTBD_RELEASE` decrements `reference_count`. Eviction eligibility restored only when `reference_count` reaches 0

5.5 Pillar IV — Kernel Driver as Hardware Control Plane

5.5.1 Why This is a Hardware Invention, Not a Software Invention

Patent offices sometimes challenge kernel driver claims as software inventions. The `kv_cpu.ko` driver is correctly characterised as a hardware control plane for the following reasons. Every user-visible interface provided by the driver — the HMAT registration, the `madvise` extensions, and the `io_uring` opcodes — terminates in a write to a hardware register in the KV-CPU silicon. The HEPC priority scoring function cannot execute correctly without receiving the decode step number via the step-advance register write. The RTBD eviction guard for prefix blocks cannot activate without receiving the `is_prefix_block` flag and reference count updates via the

RTBD_SHARE/RTBD_RELEASE register commands. The NMCE cannot locate blocks without RTBD CAM lookup. The driver software exists for one purpose: to convey LLM-semantic information (decode step, block hotness, prefix membership) from the LLM runtime to hardware that has no other mechanism to receive it. Remove the driver and the hardware policy loop cannot close. Remove the hardware and the driver is meaningless. They are co-dependent components of a single hardware-software system.

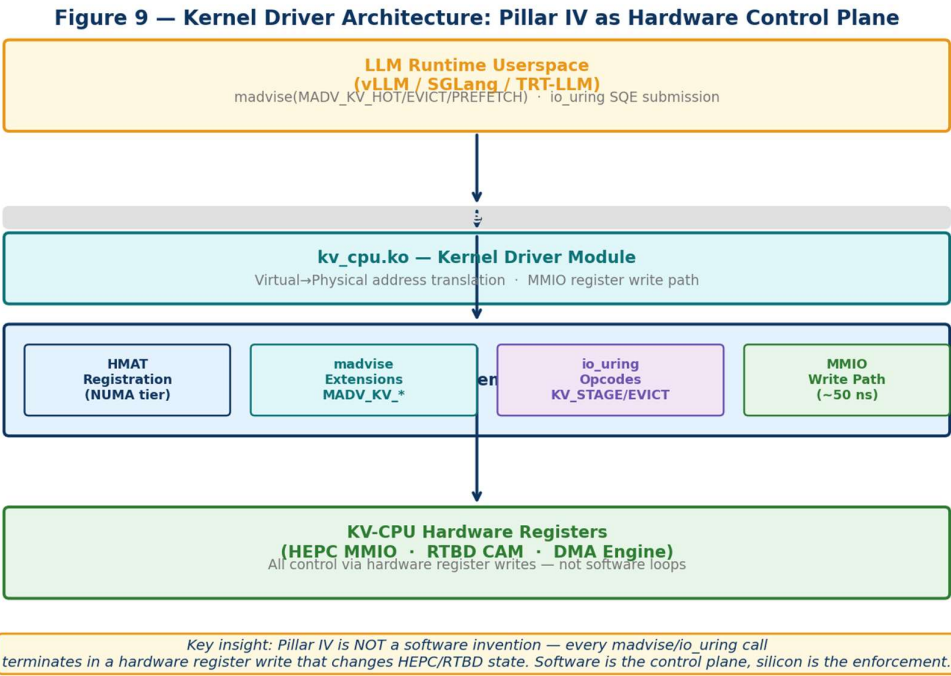


Figure 9 — Kernel Driver as Hardware Control Plane: Every API call terminates in a hardware register write

5.5.2 HMAT Memory Tier Registration

The driver registers the KV-CPU T1 LPDDR5X memory with the Linux kernel HMAT subsystem: read latency 150 ns local / 220 ns remote NUMA; write latency 155/225 ns; peak read bandwidth 200 GB/s; NUMA node ID assigned dynamically. This causes the kernel memory management subsystem to treat T1 as a first-class NUMA node addressable via mbind(), set_mempolicy(), and numactl.

5.5.3 Semantic Memory Hint Interface: Three New madvise Extensions

Hint	Arguments	Hardware Register Written	Effect on HEPC/RTBD State
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MADV_KV_HOT	addr, length	HEPC priority boost register (MMIO)	All matching RTBD entries: priority_score set to 0xFFFF. Block cannot be evicted during active decode.
MADV_KV_EVICT	addr, length	HEPC immediate-evict register (MMIO)	Matching blocks placed in eviction DMA queue immediately, bypassing normal scan phase.
MADV_KV_PREFETCH	addr, length, step, lookahead	HEPC step register + W register (MMIO)	S(B_i) recomputed; matching blocks with S > prefetch_threshold immediately scheduled for prefetch.

5.5.4 io_uring Integration for Zero-Copy Tier Migration

- IORING_OP_KV_STAGE: Async migration of KV block from T2/T3 to T1. Uses registered fixed buffers (zero-copy DMA). Completion queue entry signals GPU runtime that block is staged.
- IORING_OP_KV_EVICT: Async migration from T1 to T2 or T3. After CQE, RTBD tier_location and physical_addr fields are updated atomically by the DMA completion handler in the driver.

5.6 Minimal Embodiments — Three Independent Operating Modes

To address the requirement that each pillar be independently operable, the invention encompasses three minimal embodiments in addition to the full four-pillar device. Figure 11 illustrates these.

Figure 11 — Minimal Embodiments: Each Pillar is Independently Operable
(Embodiment A = Claim 1 only · B = Claims 5+9 only · C = All claims active)

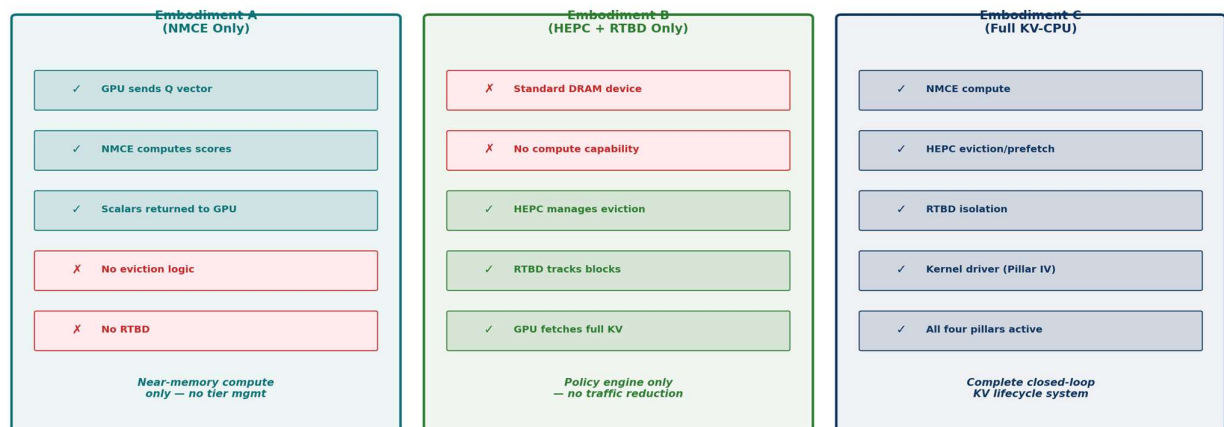


Figure 11 — Minimal Embodiments: Each Pillar Independently Operable (A: NMCE only, B: HEPC+RTBD only, C: Full device)

5.6.1 Embodiment A — NMCE-Only Device (Pillar I Alone)

A device comprising only the NMCE and its associated LPDDR5X memory, without HEPC or RTBD. The host LLM runtime manually manages which KV blocks to stage into the device's T1 memory (via standard CXL.mem writes) and manually issues NMCE compute descriptors. This embodiment still provides the 25.6x PCIe traffic reduction for attention score computation — the core benefit of Pillar I — without hardware-managed eviction. All of Pillar I claims (Claims 1–4) read on this embodiment.

5.6.2 Embodiment B — HEPC + RTBD Only Device (Pillars II + III, no Compute)

A device comprising HEPC, RTBD, and T1 LPDDR5X memory, without NMCE. This device functions as an intelligent KV-cache memory tier with hardware eviction and request isolation, but the GPU must still fetch full KV vectors across PCIe for attention computation. This embodiment eliminates the eviction latency problem (reducing from 200–600 μ s software path to ~50 ns hardware path) and provides prefix sharing, without the traffic reduction benefit of Pillar I. All of Pillar II and III claims (Claims 5–12) read on this embodiment.

5.6.3 Embodiment C — Full Four-Pillar KV-CPU

The complete device as described throughout Section 5, combining all four pillars in the closed-loop configuration. This is the preferred embodiment and is the subject of all claims.

5.7 Hardware Implementation

- Logic Die: NMCE, HEPC FSM, RTBD SRAM CAM, LPDDR5X PHY, CXL/PCIe PHY, MMIO register file, dual DMA engine. Target: TSMC N5 or N7. Estimated die area: 40–80 mm².
- Memory Chiplet: 2–4 LPDDR5X die stacks (128–512 GB). Connected via silicon or organic interposer. 4x 32-bit channels at 8.5 Gbps/pin = 256 GB/s aggregate.
- Power: Active ~20–28W; Idle ~8–12W; Standby ~3W. Form factor: PCIe HHHL or CXL EDSFF E3.S.
- Platform compatibility: Intel Xeon Scalable (Sapphire/Granite Rapids, CXL 1.1/2.0); AMD EPYC (Genoa/Turin, CXL 1.1); ARM Neoverse N2/V2; AWS Graviton 3/4; Ampere Altra Max (PCIe 5.0). Linux kernel 6.2+.

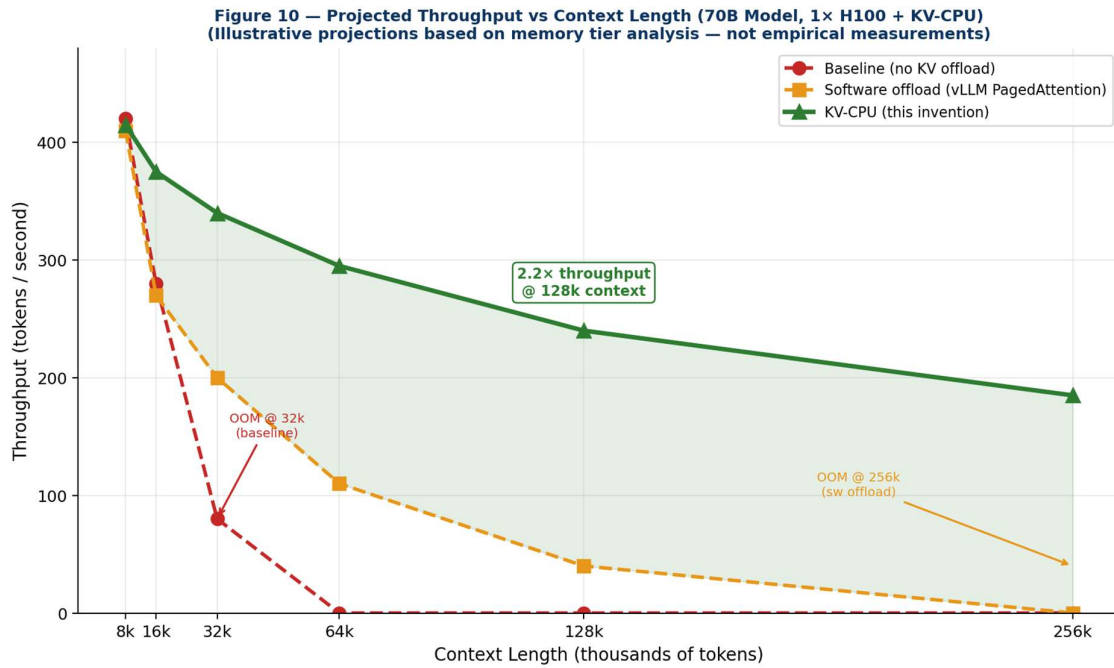


Figure 10 — Projected Throughput vs Context Length: 70B model, 1x H100 + KV-CPU vs baseline and software offload (illustrative)

6. CLAIMS

Provisional claim set under Section 9(1), Patents Act, 1970. Four-pillar architecture: Claims 1–4 (Pillar I, NMCE), Claims 5–8 (Pillar II, HEPC), Claims 9–12 (Pillar III, RTBD), Claims 13–15 (Pillar IV, Kernel Driver), Claims 16–17 (System). Each pillar has an independent claim that can be prosecuted standalone. The closed-loop combination of all four pillars is claimed in Claims 16–17.

PILLAR I — NEAR-MEMORY ATTENTION SCORE REDUCTION (Claims 1–4)

1. A semiconductor device for accelerating attention score computation in autoregressive transformer inference, the device comprising:

an on-device memory array configured to store key vectors associated with one or more transformer attention layers of one or more inference requests;
a near-memory compute engine (NMCE) comprising arithmetic circuits physically co-located on the same package as the on-device memory array, the NMCE configured to: (a) receive a query vector from an external graphics processing unit via a host interconnect, (b) access key vectors directly from the on-device memory array without transferring said key vectors across the host interconnect, (c) compute scaled dot-product attention scores between the query vector and each accessed key vector using said arithmetic circuits, and (d) transmit only the computed attention score scalar values to the external GPU via the host interconnect;
wherein the NMCE is semantically coupled to a key-value cache block lifecycle system such that the query vector received corresponds to a specific transformer layer and attention head, and the key vectors accessed are those of a specific inference request and token range as identified by a block directory maintained in hardware on the device

2. The device of claim 1, wherein the near-memory compute engine comprises: a dot-product unit array of width D equal to the attention head dimension, operable in FP16 and BF16 formats; a query vector buffer SRAM holding at least 16 simultaneous query vectors for multi-head parallel scoring; and a softmax numerator approximation unit implementing a piecewise-linear approximation of the exponential function with relative error not exceeding 0.5%, enabling attention score partial computation without softmax denominator accumulation on-device.

3. The device of claim 1, wherein the quantity of data transmitted from the external GPU to the device plus the quantity of data transmitted from the device to the external GPU per attention block score operation is substantially less than the quantity of data that would be

required to transfer the corresponding key vectors to the external GPU, the reduction factor being proportional to D/B where D is the attention head dimension and B is the number of key vectors in the block being scored.

4. The device of claim 1, wherein the attention score scalar values are transmitted to the external GPU via a peer-to-peer CXL.io or PCIe direct memory write to a memory region in the GPU's address space, without an intermediate read operation by the host CPU, and wherein a completion notification to the GPU is delivered via a completion queue entry write using the same host interconnect.

PILLAR II — DECODE-STEP-AWARE HARDWARE EVICTION (Claims 5–8)

5. A hardware eviction and prefetch controller for managing key-value cache blocks across a plurality of memory tiers in an autoregressive transformer inference system, the controller comprising:

a step-advance register accessible via a memory-mapped I/O interface, configured to receive from an external graphics processing unit a current decode step number representing the number of tokens generated in an inference session, wherein receipt of a new decode step number triggers a hardware eviction and prefetch evaluation cycle;

a priority scoring circuit implemented in fixed-function logic, configured to compute a composite priority score for each tracked key-value cache block using at least three independently weighted components: a recency component derived from time elapsed since last access measured in decode steps, a frequency component derived from cumulative access count, and a decode-step proximity component derived from the arithmetic difference between the current decode step number and the decode step number at which the block was last accessed;

an eviction circuit configured to asynchronously transfer key-value cache blocks whose composite priority scores are below a configurable threshold to a lower memory tier, without interrupting ongoing compute operations of the external GPU and without requiring execution of any instruction on the host CPU;

a prefetch circuit configured to asynchronously stage key-value cache blocks whose composite priority scores exceed a prefetch threshold from lower memory tiers into a higher-bandwidth on-device memory tier, operating on a dedicated DMA channel independent of the eviction circuit

6. The controller of claim 5, wherein the composite priority score $P(B_i)$ is computed as $P(B_i) = w_r \times R(B_i) + w_f \times F(B_i) + w_s \times S(B_i) + w_d \times D(B_i)$, wherein: $R(B_i)$ is a saturating counter initialised to a maximum value on each block access and decremented

once per received decode step number; $F(B_i)$ is a saturating counter incremented on each block access and never decremented; $S(B_i) = \max(0, W - (t_{\text{current}} - t_{\text{last_access}}))$ where W is a programmable lookahead window register and t_{current} is the most recently received decode step number; $D(B_i)$ is a binary prefix dependency flag sourced from a hardware block directory; and w_r, w_f, w_s, w_d are hardware weight registers independently configurable via the memory-mapped I/O interface.

7. The controller of claim 5, wherein the eviction circuit and the prefetch circuit each operate on separate direct memory access engine channels such that eviction of low-priority blocks to a lower tier and prefetch of high-priority blocks from a lower tier may proceed simultaneously without serialisation, and wherein completion of either operation triggers an atomic update of the physical address and tier location fields in the hardware block directory for the migrated block.

8. The controller of claim 5, wherein blocks identified as containing key-value vectors from a shared prefix common to multiple concurrent inference requests are excluded from eviction scheduling regardless of their composite priority score, the exclusion being enforced by a hardware reference counter that prevents eviction when the counter value is greater than zero.

PILLAR III — REQUEST-TAGGED MULTI-TENANT KV ISOLATION (Claims 9–12)

9. A hardware block directory for managing key-value cache memory blocks across multiple concurrently executing inference requests in a transformer inference system, the directory comprising:

a hardware tag store implemented in SRAM, each entry of the tag store associating a physical key-value cache memory block with at least: a request identifier uniquely identifying one of the plurality of concurrent inference requests; a layer index identifying a specific transformer model layer; a head index identifying a specific attention head within said layer; a token position range identifying which context tokens' key-value vectors are stored in the block; and a tier location field encoding which of a plurality of memory tiers currently stores the block's data;

a prefix sharing mechanism comprising a prefix flag field in each entry settable to indicate that the block contains key-value vectors from a context prefix shared among multiple inference requests, and a hardware reference counter per entry that is incremented when an inference request is associated with the block and decremented when an inference request releases the block, wherein the hardware eviction policy is enforced to prevent eviction of any block whose reference counter value is greater than zero;

a single-cycle lookup path that, given a tuple comprising a request identifier, a layer index, a head index, and a token position, returns the current tier location and physical address of the corresponding block using content-addressable memory hardware

10. The directory of claim 9, wherein a key-value cache block with its prefix flag set and its request identifier set to a reserved shared-prefix value is stored as a single physical copy in device memory, and wherein multiple inference requests may concurrently issue attention score compute operations against said single physical copy via the near-memory compute engine without requiring duplication of the block's data, the block's reference counter value reflecting the number of currently active requests sharing the block.

11. The directory of claim 9, wherein the tier location field is a multi-bit field encoding at least four distinct tiers comprising GPU high-bandwidth memory, device on-board memory, host CPU DRAM, and non-volatile storage, and wherein a change of tier location resulting from a hardware eviction or prefetch DMA operation causes an atomic update of both the tier location field and the physical address field in the same hardware transaction.

12. The directory of claim 9, wherein each entry further comprises an access step field storing the decode step number at which the block was most recently accessed, said access step field being written by the near-memory compute engine upon each attention score computation operation against the block, and being read by the hardware eviction and prefetch controller's priority scoring circuit to compute the decode-step proximity component of the block's composite priority score.

PILLAR IV — HARDWARE CONTROL PLANE KERNEL INTERFACE (Claims 13–15)

13. A kernel module for a host operating system, the kernel module comprising logic that, when loaded into the host operating system kernel, performs:

registering an on-device memory region of a key-value cache companion processing unit with the host operating system's memory management subsystem by creating a Heterogeneous Memory Attribute Table entry specifying access latency and bandwidth attributes of said on-device memory region, causing the operating system to assign said region a NUMA node identifier and treat it as a memory tier subject to kernel memory placement policies;

registering at least one extended memory advisory operation accessible to userspace processes via a system call, the operation accepting a virtual address range corresponding to key-value cache blocks and at least one parameter conveying semantic information about the operational state of the inference session, said semantic information selected from the group consisting of: a hotness indicator designating the blocks as currently active in an autoregressive decode loop, an

eviction indicator designating the blocks as no longer required, and a decode step number together with a prefetch lookahead window size;
upon receipt of an extended memory advisory operation, translating the virtual address range to one or more entries in a hardware block directory of the companion processing unit, and writing to one or more memory-mapped hardware registers of the companion processing unit's hardware eviction and prefetch controller to modify the eviction and prefetch scheduling state of the identified blocks, wherein no key-value cache block data is transferred to host CPU DRAM during said register write operations

14. The kernel module of claim 13, wherein the extended memory advisory operations comprise at least: a first operation causing the hardware eviction and prefetch controller to set the priority score of all blocks in the specified address range to a maximum value such that those blocks are excluded from eviction scheduling; a second operation causing the hardware eviction and prefetch controller to immediately schedule all blocks in the specified address range for background DMA eviction regardless of their current priority scores; and a third operation writing a specified decode step number to the step-advance register and a specified lookahead window size to the window register of the hardware eviction and prefetch controller.

15. The kernel module of claim 13, further registering at least two asynchronous I/O operation codes accessible via an `io_uring` kernel interface, comprising: a first operation code for migrating a key-value cache block from a lower memory tier into the companion processing unit's on-device memory using kernel-registered fixed buffers with zero intermediate copy operations; and a second operation code for migrating a key-value cache block from the companion processing unit's on-device memory to a lower tier, wherein completion of either migration atomically updates the tier location and physical address fields of the corresponding entry in the companion processing unit's hardware block directory.

SYSTEM CLAIMS — CLOSED-LOOP COMBINATION (Claims 16–17)

16. An inference server system for executing autoregressive large language model inference, comprising:

one or more graphics processing units, each equipped with high-bandwidth on-package memory, executing an autoregressive transformer decoding process wherein at each decode step a decode step number is written to a step-advance register of a companion processing unit via a memory-mapped I/O write;

a key-value cache companion processing unit connected to the graphics processing units via a CXL 3.0 or PCIe 5.0 host interconnect, the companion processing unit comprising: a near-memory compute engine as claimed in claim 1; a hardware eviction and prefetch controller as claimed in claim 5 whose priority scoring circuit receives block access step information from a hardware block directory; and a hardware block directory as claimed in claim 9 whose access step field is updated by the near-memory compute engine on each attention score computation;

a host processor executing a kernel module as claimed in claim 13;

wherein the near-memory compute engine, the hardware eviction and prefetch controller, and the hardware block directory operate as a closed-loop system wherein attention score computation operations update block access metadata, block access metadata drives eviction and prefetch scheduling, and eviction and prefetch scheduling determines which blocks are available for attention score computation, all without host CPU intervention in the critical decode path

17. The system of claim 16, wherein the key-value cache companion processing unit is fabricated as a two-chiplet semiconductor package comprising: a logic die in a semiconductor process node of 7 nm or smaller containing the near-memory compute engine, hardware eviction and prefetch controller, hardware block directory, host interconnect PHY, and direct memory access engines; and a memory die stack providing between 128 GB and 512 GB of LPDDR5X capacity, connected to the logic die via high-density interposer interconnects providing aggregate read bandwidth of at least 200 GB/s, wherein the logic die and memory die stack are co-packaged such that key vector access by the near-memory compute engine does not traverse any external interconnect.

7. ABSTRACT

A KV-Cache Companion Processing Unit (KV-CPU) is disclosed as a closed-loop AI-native memory compute architecture for large language model inference servers. The KV-CPU implements a hardware feedback loop between four co-integrated components: (1) a Near-Memory Compute Engine (NMCE) that computes dot-product attention scores between GPU-supplied query vectors and key vectors resident in on-device LPDDR5X memory — an attention-specific operation coupled to the KV block lifecycle, not generic PIM — returning only scalar scores and reducing PCIe traffic for attention scoring by a factor proportional to D/B (25.6x for D=128, B=32); (2) a Hardware Eviction and Prefetch Controller (HEPC) implementing a four-component hardware priority score (recency, frequency, decode-step proximity, prefix dependency) in fixed-function silicon, triggered by GPU MMIO step-advance signal at ~50 ns vs

200–600 μ s for software eviction, with asynchronous eviction and prefetch on separate DMA channels; (3) a Request-Tagged Block Directory (RTBD) maintaining per-block hardware metadata including request isolation and hardware reference-counted prefix block sharing, reducing prefix KV memory consumption C-fold for C concurrent requests sharing a system prompt; and (4) a Linux kernel driver functioning as the hardware control plane — registering device memory via HMAT as a NUMA tier, exposing three madvise extensions (MADV_KV_HOT/EVICT/PREFETCH) and two io_uring opcodes, each terminating in hardware register writes to HEPC or RTBD silicon state. The closed-loop combination — where NMCE output semantically informs HEPC priority, HEPC manages RTBD tier placement, and RTBD enables NMCE block location — is the central inventive concept, not present in any prior art including HBM-PIM, CXL memory expanders, or software KV eviction systems. Three minimal embodiments (NMCE-only, HEPC+RTBD-only, full device) are disclosed. The device extends effective context length from ~20k to ~256k tokens on a single H100 GPU while eliminating GPU pipeline stalls from KV cache misses.

8. DESCRIPTION OF DRAWINGS

All figures are included in this specification above. For reference:

- Figure 1 — KV-CPU System Context: Four-Tier Memory Hierarchy (T0–T3) with bandwidth/latency annotations
- Figure 2 — KV-CPU Internal Block Diagram: Logic die components and memory chiplet
- Figure 3 — NMCE Compute Pipeline: Attention-specific near-memory score reduction dataflow
- Figure 4 — PCIe Traffic Comparison: Per-block scoring without vs. with NMCE; amortised multi-head reduction
- Figure 5 — HEPC Three-Phase State Machine: Scan/Evict/Prefetch cycle and step-advance trigger
- Figure 6 — Priority Score Evolution: R/F/S/D components across 120 decode steps with eviction threshold
- Figure 7 — RTBD Entry Bit-Field Layout: ~240 bits per entry with all fields annotated
- Figure 8 — Prefix Block Sharing: Hardware reference counting vs. C-fold replication
- Figure 9 — Kernel Driver as Hardware Control Plane: Software stack terminating in hardware registers
- Figure 10 — Throughput vs Context Length: Projected performance baseline vs software offload vs KV-CPU

- Figure 11 — Minimal Embodiments: Three independently operable configurations (A: NMCE only, B: HEPC+RTBD, C: Full)
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9. STATEMENT UNDER SECTION 8, PATENTS ACT 1970

The applicant(s) declare that no corresponding application for patent for the same or substantially the same invention has been filed in any convention country as of the date of this provisional application. Should any such application be filed subsequently, the applicant undertakes to file the required statement under Section 8 within the prescribed period.

10. DECLARATION

We/I, the undersigned, being the applicant(s), hereby declare that:

19. The statements made herein are true and correct to the best of our/my knowledge, information, and belief.
20. We/I are/am the true and first inventor(s) of the invention as described herein.
21. No patent for this or substantially the same invention has been obtained or applied for in India or any convention country prior to this application.
22. We/I understand that rights under this provisional specification arise only upon filing of a complete specification within 12 months of the priority date, and that failure to do so renders this application abandoned under Section 9(1).

Signature of Applicant / Inventor

Name: _____

Date: _____

Place: _____

Signature of Registered Patent Agent (if any)

Name & Reg. No.: _____

Date: _____

Firm / Address: _____

APPENDIX A — GLOSSARY

Term	Definition
Attention mechanism	Transformer component computing a weighted sum of value vectors with weights derived from query-key dot products — the specific computation NMCE accelerates.
Autoregressive decoding	Sequential token-by-token generation where each token requires KV vectors from all previous tokens — the source of KV-cache growth.
Closed-loop KV system	The central inventive concept: NMCE compute output informs HEPC priority; HEPC drives RTBD tier placement; RTBD enables NMCE block location. All in hardware, each step.
CXL (Compute Express Link)	Open interconnect standard built on PCIe physical layer. CXL 3.0 supports peer-to-peer GPU-to-device transactions without host CPU involvement.
Decode step	One iteration of the autoregressive loop producing one token. The HEPC uses decode steps — not wall-clock time — as its eviction policy time unit.
HEPC	Hardware Eviction and Prefetch Controller. Fixed-function FSM computing R+F+S+D priority scores in silicon, not firmware.
HMAT	Heterogeneous Memory Attribute Table (ACPI). Describes memory domain latency/bandwidth to OS — used by the Pillar IV driver to register T1 as a NUMA node.
io_uring	Linux kernel async I/O interface. IORING_OP_KV_STAGE and IORING_OP_KV_EVICT are novel opcodes defined by this invention for zero-copy KV block migration.
LPDDR5X	Low-Power DDR5 Extended. 8.5 Gbps/pin. Higher capacity per cost vs HBM3; lower power than DDR5 at comparable bandwidth. The KV-CPU T1 memory medium.
NMCE	Near-Memory Compute Engine. Computes attention-specific dot-product scores using on-device key vectors. Not generic PIM — coupled to KV block lifecycle.
RTBD	Request-Tagged Block Directory. Hardware SRAM CAM providing request isolation and hardware reference-counted prefix sharing across concurrent inference requests.
Step-advance register	MMIO register written by GPU each decode step. The hardware signal that triggers the HEPC closed-loop cycle and provides transformer-temporal context to eviction policy.

APPENDIX B — FILING CHECKLIST

23. Form 1 — Application for Grant of Patent (in duplicate, signed)
24. Form 2 — Provisional Specification (this document, in duplicate)
25. Form 5 — Declaration as to Inventorship (signed by all inventors)
26. Proof of right (assignment deed if applicant is not inventor)
27. Form 26 — Power of Attorney (if patent agent appointed)
28. Official filing fee (Schedule I, Patents Act — natural person/startup rate if eligible)

Next steps: (1) Engage registered Indian patent agent (ipindia.gov.in). (2) File PCT/RO/101 within 12 months for international priority. (3) Commission formal drawings per IPO standards. (4) Conduct freedom-to-operate search against HBM-PIM, CXL expander, and vLLM prior art. (5) File complete specification with formal claim prosecution strategy within 12 months.